IT4010 Research Project – 2026 Jan

Topic Assessment Form

V2.1

Project ID: R26_IT_055

1. Topic (12 words max)

Secure Elder-Care Monitoring Platform Video Based

2. Research group the project belongs to

AIMS - Autonomous Intelligent Machines and Systems

3. Specialization of the project belongs to

Information Technology (IT)

4. If a continuation of a previous project:

Project ID	
Year	

5. Brief description of the research problem including references (300 words max) – references not included in word count.

The rapid growth of the global elderly population has significantly increased dependence on in-home caregivers, creating serious concerns regarding elder safety, caregiver accountability, and continuous supervision. Elderly individuals who live alone or suffer from reduced mobility are particularly vulnerable to falls, prolonged inactivity, and unsafe or negligent caregiver behavior. Despite the widespread use of home-care services, families and care institutions lack reliable, non-intrusive mechanisms to continuously monitor caregiver activities and detect abnormal or unsafe behavior in real time, especially within private home environments where privacy must be preserved.

The primary stakeholders affected by this problem include elderly individuals, family members, home-care agencies, and healthcare service providers. Elderly users face increased physical risk due to delayed fall detection, neglect, or violent behavior, which may result in severe injuries or even fatalities. Family members experience emotional stress, anxiety, and reduced trust in caregiving services due to limited transparency. Care agencies are exposed to reputational damage, legal liability, and regulatory challenges arising from the absence of objective monitoring and behavioral evidence.

The impact of this problem spans multiple dimensions. Socially, delayed detection of critical incidents directly threatens the safety and well-being of elderly individuals. Technically, existing monitoring solutions often rely on wearable devices or cloud-based video analytics, which suffer from poor user compliance, privacy concerns, bandwidth dependency, and high response latency. Economically, delayed medical intervention increases healthcare costs for families and institutions. Ethically and legally, continuous video surveillance in private homes raises serious privacy and data protection concerns. Environmentally, cloud-intensive systems contribute to higher energy consumption compared to edge-based processing solutions.

This research is positioned within the Artificial Intelligence and Computer Vision research cluster and targets the Healthcare Technology and Smart Surveillance industry verticals. By employing privacy-preserving, pose-based anomaly detection on edge devices, the research supports the achievement of United Nations Sustainable Development Goals, particularly SDG 3 (Good Health and Well-Being) and SDG 11 (Sustainable Cities and Communities), by enabling safer, trustworthy, and sustainable elder-care environments

[01] Development of an IoT-Based Smart Elder Care System KDU Int. Research Conf., General Sir John Kotelawala Defence Univ., Sri Lanka, 2022.
Link:https://ir.kdu.ac.lk/bitstream/handle/345/6472/IRC%202022%20Proceedings%20_Com_draft%20FOC-415-421.pdf

[02] LighterFace Model for Community Face Detection and Recognition MDPI, 2024
Link:https://www.researchgate.net/publication/379845123_LighterFace_Model_for_Community_Face_Detection_and_Recognition

03] Anomaly Detection in Surveillance Camera Data Bachelor's Thesis, Lund University, Sweden, 2023.https://bth.diva-portal.org/smash/get/diva2:1774131/FULLTEXT02.pdf

[04] Video-Based Person Re-Identification: Methods, Datasets, and Deep Learning Int. J. Engineering Advanced Technology (IJEAT), vol. 9, no. 3, 2020. Link: https://www.ijeat.org/wp-content/uploads/papers/v9i3/C6524029320.pdf

[05] ArcFace: Additive Angular Margin Loss for Deep Face Recognition ResearchGate | CVPR 2019 Link:
https://www.researchgate.net/publication/335173501_ArcFace_Additive_Angular_Margin_Loss_for_Deep_Face_Recognition

[06] J. Deng et al., "ArcFace: Additive Angular Margin Loss for Deep Face Recognition," in Proc. IEEE/CVF Conf. Computer Vision Pattern Recognition (CVPR), Long Beach, CA, USA, 2019, pp. 4690-4699, doi: 10.1109/CVPR.2019.00453. [Online]. Available: https://ieeexplore.ieee.org/document/8954101

[07] Census of Population and Housing 2024, Central Bank of Sri Lanka, Oct. 2024. [Online].
Available: https://www.statistics.gov.lk/qlink/PressReleases/CPH2024_Press_Release_Eng

[08] LighterFace Model for Community Face Detection and Recognition MDPI, 2024 Link:
https://www.researchgate.net/publication/379845123_LighterFace_Model_for_Community_Face_Detection_and_Recognition

6. Brief description of Existing Research and Systems (300 words max)

Elder Care Applications:

IoT Smart Elder Care (KDU Sri Lanka 2022): Sensor-based activity monitoring. **Gap**: No video analytics; privacy-preserving edge processing missing.

Meal Monitoring Sensors (MDPI 2022): Weight/temperature sensors. **Gap**: Narrow scope; ignores caregiver behavior.

Computer Vision & Tracking:

- **ArcFace (IEEE CVPR 2019)**: State-of-the-art face recognition (99.8% accuracy). **Gap**: Not optimized for edge devices or real-time elder care.
- **ByteTrack (ECCV 2022)**: Multi-object tracking benchmark. **Gap**: General surveillance; no elder-care behavioral context. https://ieeexplore.ieee.org/document/10947826
- **CCTV Physiological Monitoring (IEEE 2024)**: Heart rate, breathing rate from cameras. **Gap**: Clinical focus; no caregiver accountability. https://ieeexplore.ieee.org/document/10229220

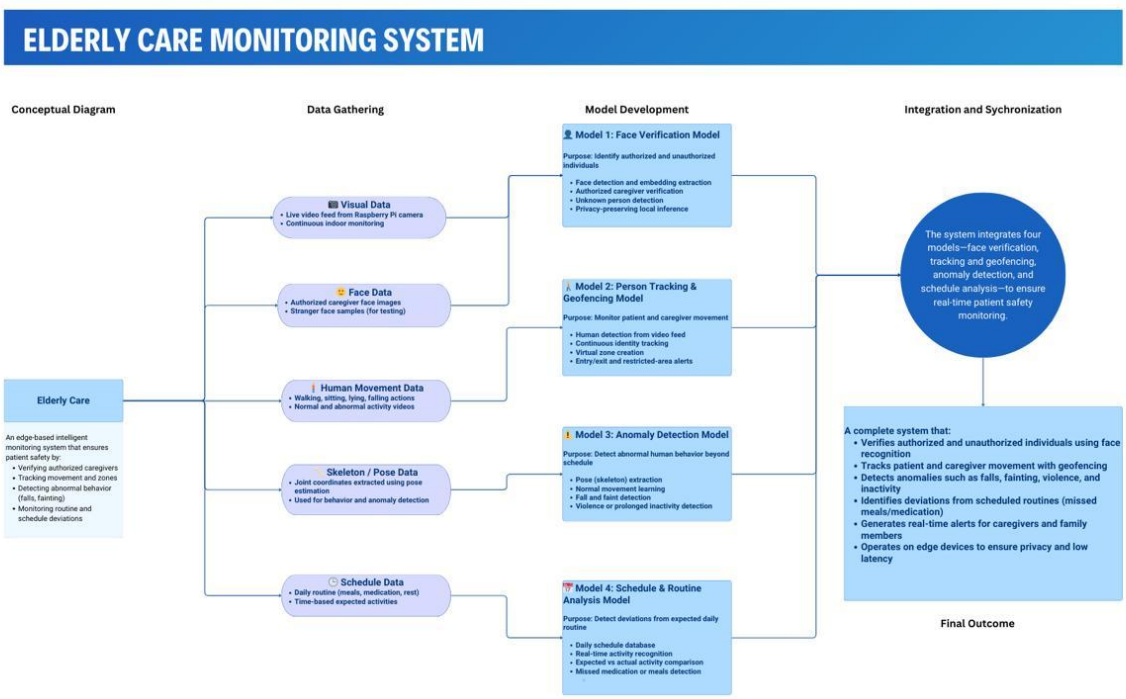
7. Brief description of the solution's nature, including a conceptual diagram (maximum 500 words).

The proposed solution is an AI-driven, privacy-preserving elder-care monitoring system designed to enhance patient safety, caregiver accountability, and real-time incident detection within home-care environments. The system addresses key limitations of existing solutions, including privacy risks, high latency, dependence on wearable devices, and limited behavioral intelligence. By utilizing edge-based computer vision and pose-based behavior analysis, the system enables continuous monitoring without transmitting raw video data to the cloud.

To overcome the identified gaps, the solution integrates four specialized AI models operating collaboratively on a Raspberry Pi edge device. All sensitive processing is performed locally, reducing privacy concerns and ensuring low-latency response. The use of skeleton-based pose data instead of raw video frames enables ethical monitoring while preserving essential motion characteristics. Beyond conventional fall detection, the system incorporates caregiver tracking, virtual geofencing, and schedule deviation analysis, providing a holistic monitoring framework.

From a system perspective, the solution follows an input–process–output architecture. Inputs include **live video feeds captured**, facial image data for caregiver verification, skeletal joint coordinates extracted using pose estimation, and predefined schedule data representing daily routines such as meals and medication times. The processing layer consists of multiple AI modules: a face verification model to authenticate caregivers, a person tracking and geofencing module to monitor movement, a pose-based anomaly detection model to identify falls, violence, or prolonged inactivity, and a schedule analysis module to detect deviations from expected routines.

The system outputs real-time alerts, structured event logs, and evidence snapshots, which are securely delivered to family members and administrators. Overall, the proposed solution provides a scalable, low-latency, and privacy-aware approach that advances intelligent elder-care monitoring through ethical AI and edge computing.



8. Brief overview of the availability of necessary specialized domain expertise, knowledge, and data. (500 words max)

The successful execution of the proposed AI-driven elder-care monitoring system requires multidisciplinary expertise spanning computer vision, machine learning, edge computing, and healthcare-oriented system design. The project team can acquire essential domain knowledge through a combination of academic literature review, online professional learning platforms, and supervised experimentation. Peer-reviewed research articles from IEEE Xplore, Google Scholar, and ResearchGate provide foundational understanding of pose estimation, anomaly detection, and intelligent surveillance systems. Additionally, structured courses available on platforms such as LinkedIn Learning and Coursera offer practical insights into computer vision, deep learning, and IoT-based system development.

Technical knowledge and skills required for the project can be developed through hands-on implementation and experimentation. Team members will gain proficiency in Python programming, OpenCV, MediaPipe Pose, and deep learning frameworks such as TensorFlow and PyTorch by building and testing individual system components. Experience with edge computing will be developed through deploying optimized machine learning models on Raspberry Pi devices. Version control tools, containerization technologies, and backend frameworks such as FastAPI will further support system integration and collaborative development. Regular consultations with the project supervisor and iterative prototyping will strengthen technical competence and research rigor.

The data required for the project will be collected through controlled and ethical means. Video data capturing normal and abnormal human activities will be recorded using cameras in simulated home environments with the informed consent of participants. To preserve privacy, raw video footage will not be stored or shared externally; instead, pose estimation techniques will be applied to extract skeletal joint coordinates, which are then stored as numerical time-series data for model training. Publicly available datasets related to fall detection and human activity recognition may also be used for benchmarking and validation purposes.

This project involves human activity data and therefore requires ethical consideration. Ethical clearance will be sought in accordance with institutional guidelines to ensure informed consent, data minimization, anonymity, and responsible data handling. No personally identifiable information will be retained, and all processing will prioritize privacy preservation, making the research compliant with ethical and legal standards for human-centered AI systems.

9. Objectives and Novelty

Main Objective

To design and develop a privacy-preserving, AI-driven elder-care monitoring system that utilizes edge-based computer vision and machine learning techniques to verify caregivers, track movements, and detect behavioral anomalies in real time, thereby improving elder safety, caregiver accountability, and trust in in-home care environments.

Member Name with Registration No	Sub Objective	Tasks	Novelty
JAYASINGHE S M B H J IT22240392	To develop a real-time, edge-based facial verification system that authenticates caregiver identity while preserving privacy in an elder-care environment.	• Set up the Raspberry Pi 4 environment with a 64-bit operating system optimized for computer vision tasks • Integrate the Pi Camera and configure real-time video frame capture • Install and configure Insight Face / Arc Face models with ONNX Runtime for efficient edge inference • Collect and preprocess facial images of authorized caregivers under different lighting conditions	• Performs fully offline, edge-based facial verification, eliminating dependency on cloud services • Uses face embeddings instead of raw images, enhancing privacy and data security • Optimized for low-power hardware using ONNX Runtime and lightweight deep learning models • Applies a threshold-based similarity decision system that is transparent and explainable

		• Generate facial embeddings and store them securely in a local .npy databas • Implement real-time face detection and embedding extraction logic • Compare live face embeddings with stored authorized embeddings using similarity scoring • Define a verification threshold to classify individuals as "Authorized" or "Unknown" • Trigger access logs and alerts when an unknown person is detected Ensure raw video data is not stored or transmitted, preserving user privacy	• Minimizes sensitive data storage by saving images only during unauthorized access events Enhances elder safety through real-time identity verification at the point of entry
IT22171788	Monitors in real-time whether a person enters or exits a defined radius around a house.	Download & setup YOLOv8-Nano model weights + Ultralytics library on Pi. Integrate ByteTrack tracker to maintain consistent person IDs during movement. Build GUI tool for users to click-draw polygon zones and save to zones.json.	Edge-Native: All processing happens locally on Pi—no raw video sent to cloud (privacy-preserving). Superior Accuracy: ByteTrack achieves 77.3% MOTA. • Smart Tracking: Only 558 ID switches per video, ensures

		Write logic to check if person coordinates fall inside restricted zone polygons. • Test end-to-end pipeline: YOLO detection → ByteTrack tracking → Zone violation checking.	person identity stays consistent.
IT22075208 Rathnayaka R M V R	To design and implement a real-time, pose-based anomaly detection module that identifies falls, violent actions, and prolonged inactivity of caregivers, while preserving privacy through edge-based processing.	1. Collect and record video datasets representing both normal (walking, sitting, bending) and abnormal (falling, aggressive movements, inactivity) human behaviors in controlled environments. 2. Apply MediaPipe Pose to extract skeletal joint coordinates (x, y) from video frames and convert them into time-series motion data. 3. Preprocess and normalize skeleton data to ensure robustness against variations in camera angle, scale, and movement speed. 4. Design and train lightweight machine learning	he novelty of this component lies in the implementation of a **privacy-preserving, skeleton-based anomaly detection system deployed entirely on edge hardware for elder-care monitoring**. Unlike existing research that primarily focuses on fall detection in controlled environments or relies on cloud-based processing, this work extends anomaly detection to include caregiver aggression and prolonged inactivity. Additionally, the system uses only skeletal motion data instead of raw video, reducing privacy risks while maintaining behavioral accuracy. The integration of lightweight temporal models optimized for

		models such as LSTM networks or Autoencoders to learn normal movement patterns. 5. Develop anomaly scoring logic based on reconstruction error or temporal deviation to classify abnormal behaviors. 6. Optimize and integrate the trained anomaly detection model for real-time execution on Raspberry Pi edge devices. 7. Validate the model performance using accuracy, false-positive rate, and detection latency metrics.	real-time edge deployment in private home environments represents a unique and practical contribution not commonly addressed in existing elder-care monitoring research.
CHAMATHKA.BMGN IT22143686	To develop a system that monitors daily activities and detects deviations from predefined schedules to enable early intervention.	• Design the daily schedule and routine database • Define rule-based conditions for each scheduled activity • Receive identity, location, and behavior data • Compare detected actions with scheduled expectations • Detect schedule deviations such as missed meals or medication	• Separates **physical anomaly detection** from **schedule compliance checking** • Uses explainable, rule-based logic • Integrates multi-model outputs into a single decision layer • Enables early intervention through routine deviation detection

		• Generate routine-based alerts and reports	

10. This part is to be filled by the Supervisor and the Co-supervisor of the Project.

a) This research topic possesses a comprehensive scope suitable for a final-year project.

Yes	x	No	

b) The proposed topic exhibits novelty.

Yes	x	No	

c) The student group can successfully execute the proposed project.

Yes	x	No	

d) The proposed sub-objectives reflect the students' areas of specialization.

Yes	x	No	

e) Any other comments:

	Title	First Name	Last Name	Signature
Supervisor	Dr	Dharshana	Kasthurirathna	
Co-Supervisor	Mrs	Chathurya	Kumarapperuma	
External Supervisor				
Summary of external supervisor's (if any) experience and expertise				